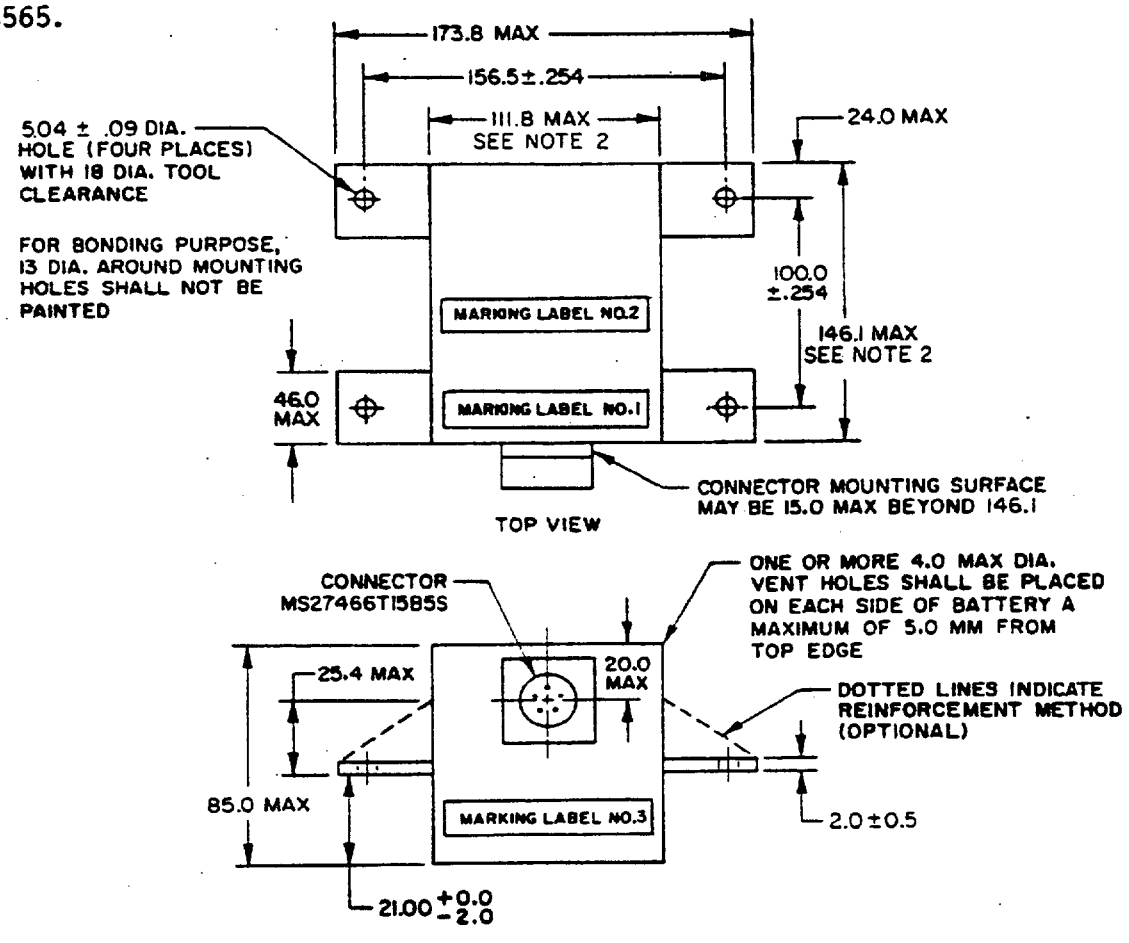
SPECIFICATION SHEET

BATTERY, STORAGE, AIRCRAFT, MEDIUM-RATE, TYPE 1,
MAINTENANCE-FREE, 24-VOLT, 1.5 AMPERE-HOUR

This specification sheet is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the storage battery described herein shall consist of this document and the latest issue of Specification MIL-B-8565.



NOTES: 1. All dimensions are in millimeters.
2. Allow additional 4 mm width for cover.

FIGURE 1. Dimensions P/N D8565/6-1.

AMSC N/A FSC 6140

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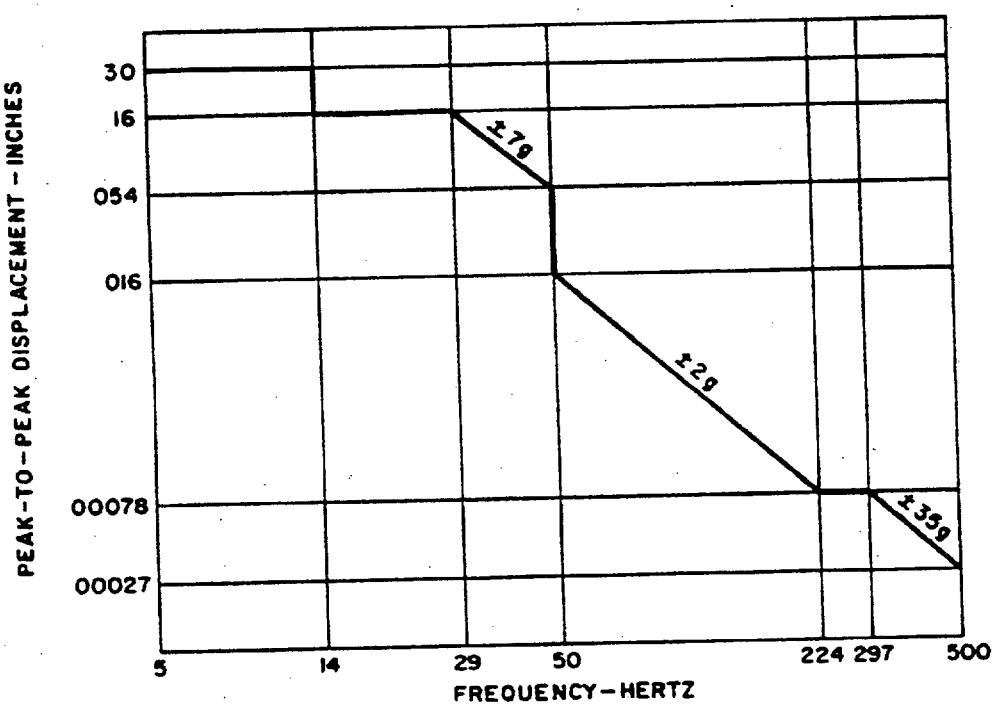


FIGURE 2. Sine wave vibration test levels (4.6.23).

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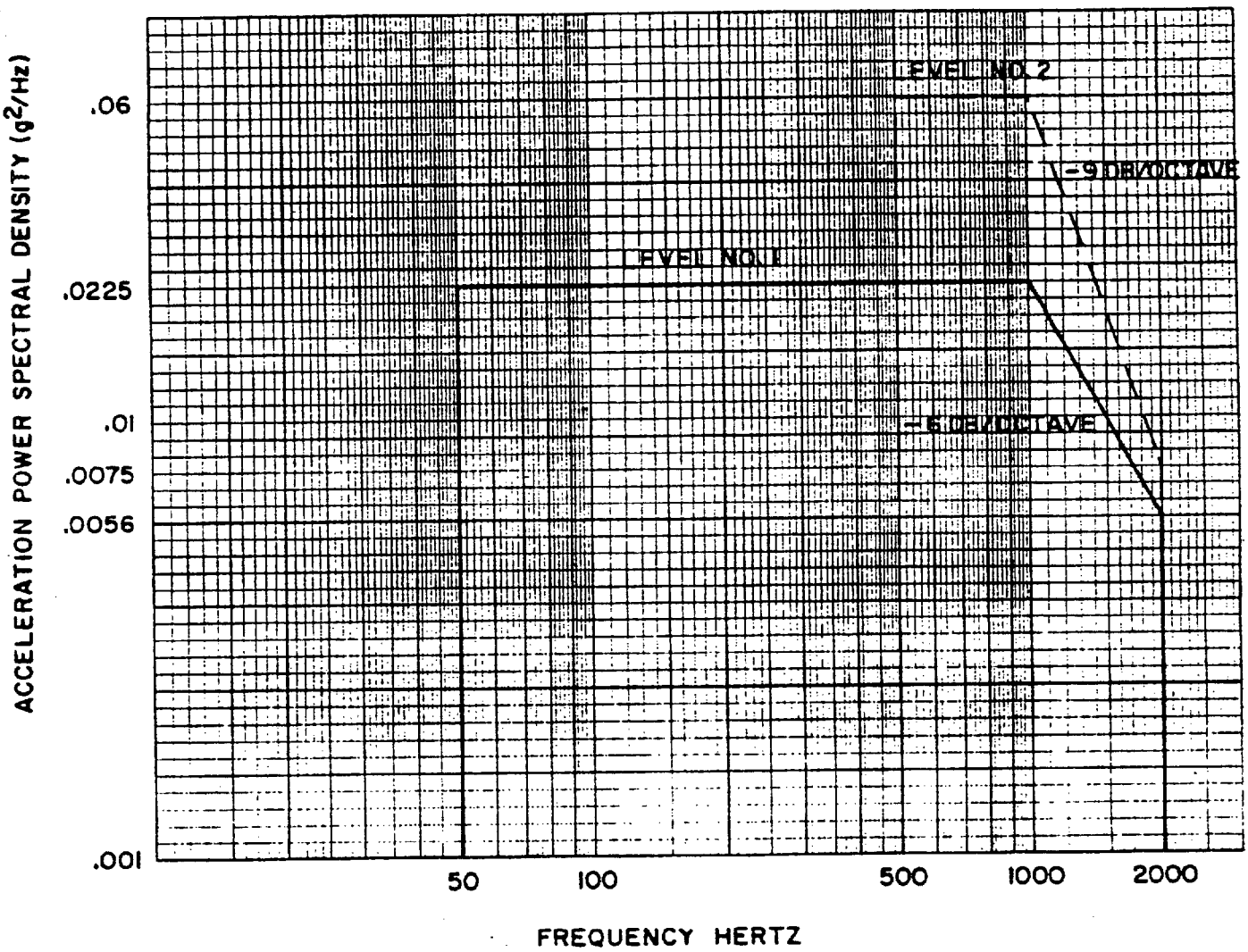


FIGURE 3. Random vibration test levels (4.6.23).

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REQUIREMENTS:

1. The dimensions of the battery shall be as shown in figure 1.

2. The military part number of the battery shall be D8565/6-1.

3. The rated capacity of the battery shall be 1.5-ampere-hour (1.5 Ah/1 HR/21°C/18.0V at end of life). 1C = 1.5 amperes.

4. The weight of the battery shall not exceed 2.9 kg.

5. Aircraft installation. For installation in the aircraft, the battery shall be supported as shown in figure 1 of MIL-B-8565. Vibration isolation may be provided to insure that the vibration level does not exceed the requirements of the 3-year accelerated life tests of 4.6.13. If vibration isolators are used, sufficient sway clearance between the battery and the mounting structure shall be provided. Battery life in the aircraft shall be 3 years.

6. The battery shall conform to military specification MIL-B-8565 for type 1 batteries with the following additions and deletions.

   3.5.4 Venting. Delete and add the following: "Each battery shall include two vent holes, located as shown in figure 1. The battery vent holes shall be made so that air can enter the battery through either vent hole, flow freely across the cells, and exit through the other vent hole. No corrosive, ignitable, or liquid discharge through the battery vent holes shall occur during testing to this specification (except when testing to special test 4.6.22). With the battery vent holes open and the battery operating within design requirements of this specification, cell venting shall not cause corrosive or liquid discharge from the battery. After a venting sequence, the venting mechanism within the battery shall automatically revert to its pre-vented configuration. Venting shall not be toxic."

   3.5.5 Receptacles. Add: "The receptacle shall be MS27466T15B5S.

   a. Pin A shall be +28 VDC input for battery heater and optional charging.

   b. *Pin B shall be charging diode output.

   c. *Pin C shall be battery charge input.

   d. Pin D shall be battery positive output.

   e. Pin E shall be battery negative.

   *Pins B and C of mating connector in aircraft installation may be jumpered if the optional charging from Pin A is used."

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3.6.3.2.1 Polarity marking. Delete and add: "The following shall be marked on label 3 on figure 1 of the specification sheet.

CONNECTION PIN IDENTIFICATION

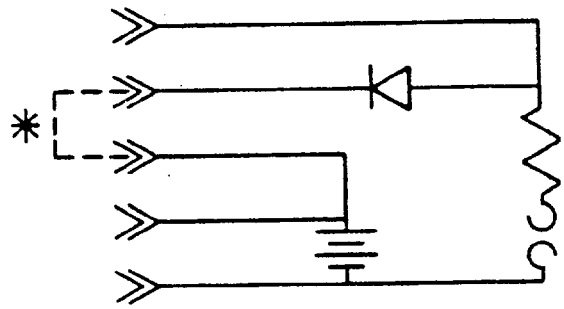
PIN A +28 VDC

PIN B CHG. DIODE

PIN C CHG. POS.

PIN D BATT. POS.

PIN E BATT. NEG.



*Pins B and C of mating connector in aircraft may be jumpered.

If heaters are not incorporated in the battery, the words "NO HEATER USED" shall be substituted in place of the heater circuit."

3.6.3.2.3 Battery caution marking (type 1 battery only). Change "0.25 inch" and "0.375 inch" wherever they occur to "0.15 inch."

3.6.7 Vent tubes. Delete.

3.6.8 Capacity and electrical performance. Delete "Requirement Number (4)."

3.6.9 Strength of receptacle. Delete.

3.6.10.1 Life cycling (type 1 battery). Change "1-hour rate discharge requirement" to "2-minute discharge at 15 amperes (18.0 volts minimum) requirement."

Add the following paragraphs:

3.6.23 Battery output performance.

3.6.23.1 Emergency loads. Batteries, when discharged in accordance with the load profile sequence shown below, shall maintain a terminal potential of 18 volts or greater.

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TABLE I

Step	Disch. Temp. (°C)	Load Wattage (Watts)	Discharge Time (Seconds)
1	-29	360 250 Rest Open Circuit 360 250	0.20 30 60 0.20 30
2	50	360 250 Rest Open Circuit 360 250 Rest Open Circuit 360 250	0.20 60 60 0.20 60 60 0.20 60
3	0	360 250 Rest Open Circuit 360 250 Rest Open Circuit 360 250	0.20 45 60 0.20 45 60 0.20 45

3.6.24 Cold and hot temperature charging.

3.6.24.1 Half-hour charging. The battery having a 0.75 Ah capacity at temperatures in the range of -18°C to 55°C shall be capable of delivering 1.25 Ah capacity after a 28.25 ± 0.5V charging source has been applied to the battery for 30 minutes.

3.6.24.2 One-hour charging. The battery, when discharged at temperatures in the range of -40°C to -17°C at 1.0A to 18.0V, shall be capable of delivering 1.25 Ah capacity after a 28.25 ± 0.5V charging source has been applied to the battery for 1.0 hour. The battery shall not be damaged when a 28.25 ± 0.5V charging source is applied continuously to the battery at -40°C.

3.6.24.3 Heaters. Heaters may be incorporated into the battery to meet the cold temperature charging requirements. The heaters shall utilize 28.25 VDC power in accordance with MIL-STD-704, accessible at connector pin A. Heater current shall not exceed 2A at 28.25 VDC. Heater power shall be provided during all constant-potential charges.

3.6.24.3.1 The heater element shall be fabricated and installed in the battery assembly using insulated heaters and wire assemblies so as to minimize the possibility of sparking.

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3.6.24.3.2 The battery shall contain temperature limiting devices as needed to prevent overheating when the heater circuit is energized. Under any condition of operation specified herein, the battery case temperature shall not exceed 60°C due to heater and battery operation. If the battery case temperature exceeds 49°C, caution labels stating: "CAUTION HOT SURFACE" shall be attached to the battery cover and to the surface bearing the battery identification. Both temperature conditions above apply at an ambient temperature of 25° ± 5°C. The heater control devices shall be compatible with the battery heater current requirements. If thermal switches are used, they shall be a snap-action type, in accordance with MIL-S-24236. The switches shall be designed and installed to prevent ignition of an explosive mixture of hydrogen and oxygen. All components of the heater circuit shall be located inside the battery case. In the event of primary heater control failure, the battery shall not reach a damaging temperature.

4.6.7 Strength of vent tubes. Delete.

4.6.11 Constant voltage discharge (14.0 volts). Delete.

4.6.13 Life. Delete and add the following: "The battery shall be subjected to 85 cycles of discharge and charge. Each cycle shall consist of a 2-minute discharge at 15 amperes, followed immediately by a 118-minute charge, followed by a 1-hour rest period. The battery shall meet the specified requirements of 3.6.10.1."

4.6.15 Charge and discharge test at low temperature. Delete.

4.6.20 Special tests. Add the following:

4.6.20.1 Battery output performance test.

4.6.20.1.1 Emergency loads. Battery samples 1, 2 and 3 shall be used to demonstrate the performance requirements of 3.6.23.1. The tests shall be conducted as follows:

a. Charge each battery at 28.25 ± 0.5V for 5 hours at 25°C.

b. Discharge samples as follows:

(1) Sample 1 in accordance with step 1 of 3.6.23.1.

(2) Sample 2 in accordance with step 2 of 3.6.23.1.

(3) Sample 3 in accordance with step 3 of 3.6.23.1.

c. Charge each battery at 28.25 ± 0.5V for 5 hours at 25°C.

4.6.20.2 Cold and hot temperature charge test.

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4.6.20.2.1 Half-hour charging. Battery samples 1, 2 and 3 shall be used to demonstrate the cold and hot temperature charging requirements of 3.6.24. The test shall be conducted as follows.

a. Charge each battery at 28.25 ± 0.5V for 5 hours at 25°C.

b. Stabilize sample 1 at -18°C, sample 2 at 15°C and sample 3 at 55°C, then discharge at 1.5A to 18.0V. Each battery shall provide at least 1.25 Ah minimum.

c. Stabilize each battery at 25°C, then charge at 28.25 ± 0.5V for 5 hours.

d. Stabilize each battery at -18°C, 15°C and 55°C, respectively, then discharge at 1.5A to remove all but .75 Ah from batteries, as determined by 4.6.20.2.1b.

e. Rest battery to stabilize at test temperatures, then apply 28.25 ± 0.5V for 30 minutes.

f. Immediately following charge, remove batteries from test chambers and discharge at 1.5A to 18.0V. Each battery must deliver 1.25 Ah minimum.

4.6.20.2.2 One-hour charging. Battery sample 3 shall be subjected to a -40°C charge demonstration test and shall be conducted as follows:

a. Charge the battery at 28.25 ± 0.5V for 5 hours at 25°C.

b. Stabilize the battery at -40°C, and then discharge at 1.5A to 18.0V.

c. Rest battery to stabilize at -40°, then apply 28.25 ± 0.5V for 1.0 hour.

d. Immediately following charge, remove battery from test chamber and discharge at 1.5A to 18.0V. Battery shall deliver 1.25 Ah minimum.

4.6.23 Vibration. Delete and add the following: "The battery assembly, including connector and cable, shall be mounted to the test apparatus, per 4.5.5 and figure 1, and vibrated along each orthogonal axis, per figure 1, in accordance with the test levels of figure 2 of the detail specification sheet. Two 15-minute logarithmic sweeps shall be conducted from 5-500-5 Hertz and one 5-minute logarithmic sweep will be conducted from 20-30-20 Hertz. A resonance search shall be conducted from 5-500 Hertz and resonance dwell testing performed for 1.5 minutes at each resonance, to a maximum of three most severe exhibiting a transmissibility of two or greater. The battery shall also be subjected to two separate random vibration tests along each orthogonal axis; one for a period of 20

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minutes per axis to level No. 1 of figure 3 of the detail specification sheet, and the second for a period of 5 minutes per axis to Level No. 2 of figure 3 of the detail specification sheet. The battery heaters shall be energized, and the battery charged at 28.25 volts during vibration, and shall be discharged at 15.0 amperes for 1.0 minute prior to beginning each axis of vibration."

Preparing Activity:
Navy - AS
Project No. 6140-N623

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TABLE 1. Inspection of batteries and order of test.

Inspection Number	Inspections	Qualification Inspection			Requirements Paragraph	Method of Inspection Paragraph	Quality Conformance Inspection		
		Qualification Sample Number					Group A Tests	Group B Tests	
		1	2	3				1	2
1	Visual & Mechanical Examination	X	X	X	3.5, 3.6.2.2 & 3.6.3	4.6.3	X	X	X
2	Dimensions & Weight	X	X	X	3.6.5	4.6.4	X	X	
3	Dielectric Strength	X	X	X	3.6.4	4.6.5	X	X	X
4	Color and Marking	X	X	X	3.6.3	4.6.8	X	X	X
5	Conditioning Charge	X	X	X	4.5.1.4	4.6.9		X	X
6	Capacity Discharge	X	X	X	3.6.8	4.6.10	X	X	X
7	Battery Output Performance	X	X		3.6.23	4.6.20.1			
8	Cold Temperature Charge	X	X	X	3.6.24	4.6.20.2			
9	Life Cycling	X			3.6.10	4.6.13		X	
10	Capacity Discharge at 49°C		X		3.6.8	4.6.14			
11	Discharge While Inverted		X		3.5.8 & 3.6.8	4.6.16			X
12	Altitude		X		3.6.11 & 3.6.12	4.6.17			X
13	Mechanical Shock	X			3.6.11 & 3.6.12	4.6.18			
14	Temperature Shock	X			3.6.11 & 3.6.12	4.6.19		X	
15	Temperature Rise & Float		X		3.6.13	4.6.21			X
16	Battery Gas Emission Test		X		3.6.14	4.6.22			
17	Vibration		X		3.6.16	4.6.23			X
18	Humidity		X		3.6.17	4.6.24			X
19	Salt Fog	X			3.6.18	4.6.25		X	
20	Ground Storage			X	3.6.15	4.6.26			
21	Physical Integrity at High Temperature	X			3.6.12	4.6.27			
22	Final Examination	X	X	X	3.6.4	4.6.30		X	X

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